Fine-grained multi-prompt essay scoring with multi-level disentanglement

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Introduction

- Automated Essay Scoring (AES) has recently gained significant attention with the rapid advances of language models, showing strong results especially in single-prompt setups.
- However, most existing approaches remain limited to single-prompt or single-trait evaluation, which does not reflect the diverse and complex scenarios of real-world education.
- In practice, students are asked to write on a variety of prompts, and each prompt is evaluated across multiple traits such as content, organization, grammar, and style.
- Even when traits share the same name, their evaluation criteria can differ depending on the prompt, highlighting the need for fine-grained modeling.

Introduction

- Even when traits share the same name, like Content, their evaluation criteria can vary greatly depending on the prompt.

- Treating them as identical leads to inaccurate and oversimplified scoring.

[Content Rubric of P1, P2, P8]
This property checks for the amount of content and ideas present in the essay.
Score 6: The writing is exceptionally clear, focused, and interesting. It holds the reader's attention throughout. Main ideas stand out and are developed by strong support and rich details suitable to audience and purpose. …

[Content Rubric of P5, P6]
Score 4: The response answers the question asked of it. Supporting evidence is specific to the memoir is used to support the points the writer makes. …

[Content Rubric of P7]
Score 3: Tells a story with ideas that are clearly focused on the topic and are thoroughly developed with specific, relevant details. …

Fig. 1 The evaluation criteria for *content* trait vary depending on the prompts

Introduction

- Embeddings of the same trait across different prompts often collapse into overlapping distributions.
- This collapse causes prompt-specific information to be lost, making fine-grained disentanglement essential.

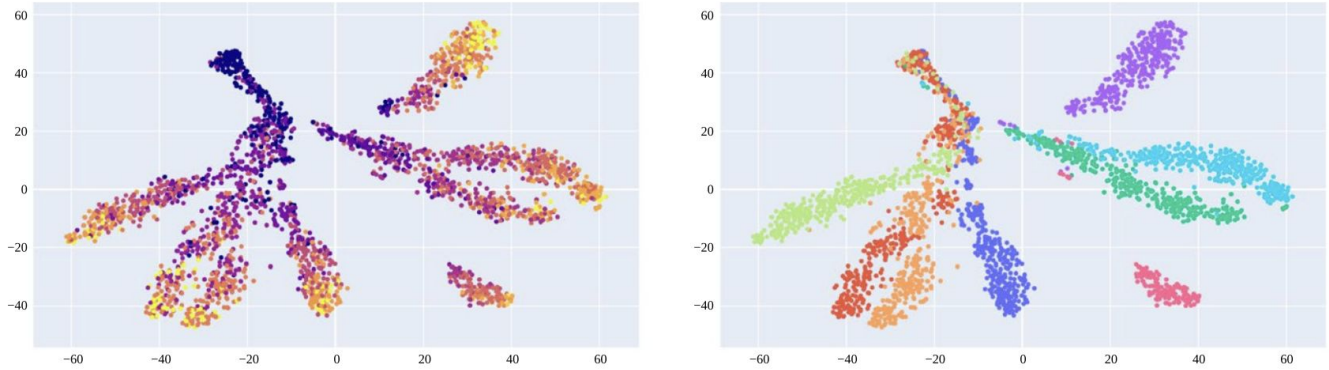


Fig. 2 T-SNE visualisation of the embeddings extracted from the language model trained to evaluate *content* scores. (Colors: Left-Score, Right-Prompt)

Contribution

- We propose a novel multi-level disentanglement framework MLPAS for multi-prompt and multi-trait essay scoring, enabling fine-grained disentanglement at both prompt and trait levels.
- MLPAS demonstrates the highest agreement with human essay ratings compared to existing state-of-the-art multi-prompt essay scoring methods.
- MLPAS outperforms existing state-of-the-art essay scoring methods even in cross-prompt setups.
- We experimentally demonstrate that our method achieves superior performance while utilizing fewer parameters and exhibiting lower latency compared to the most recent method.

Proposed Method

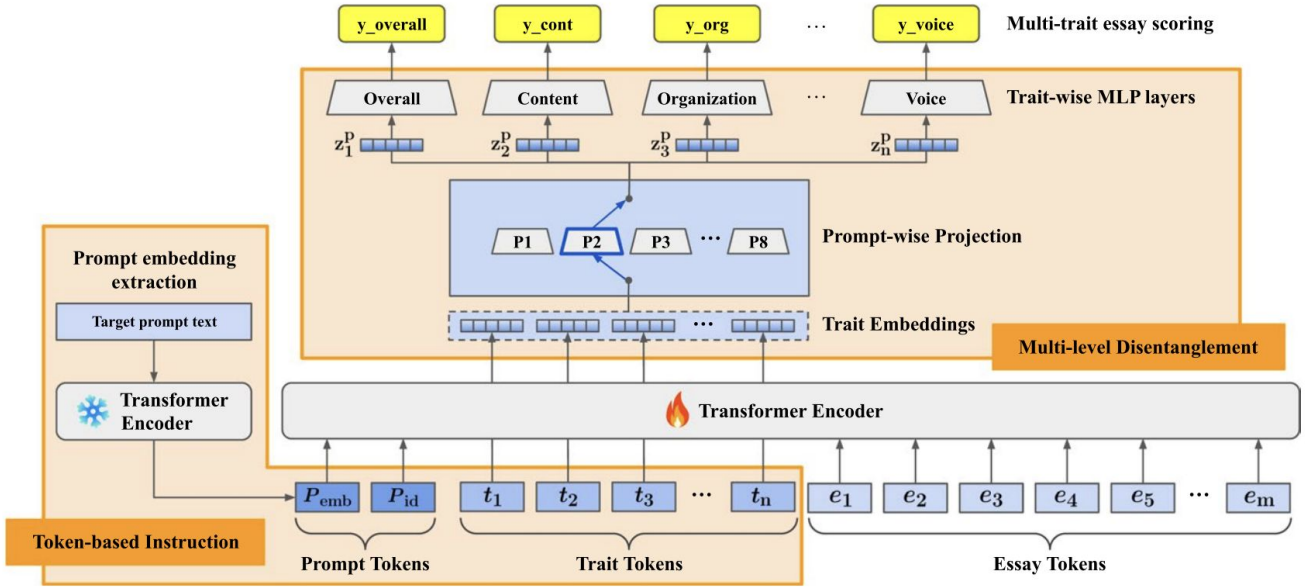
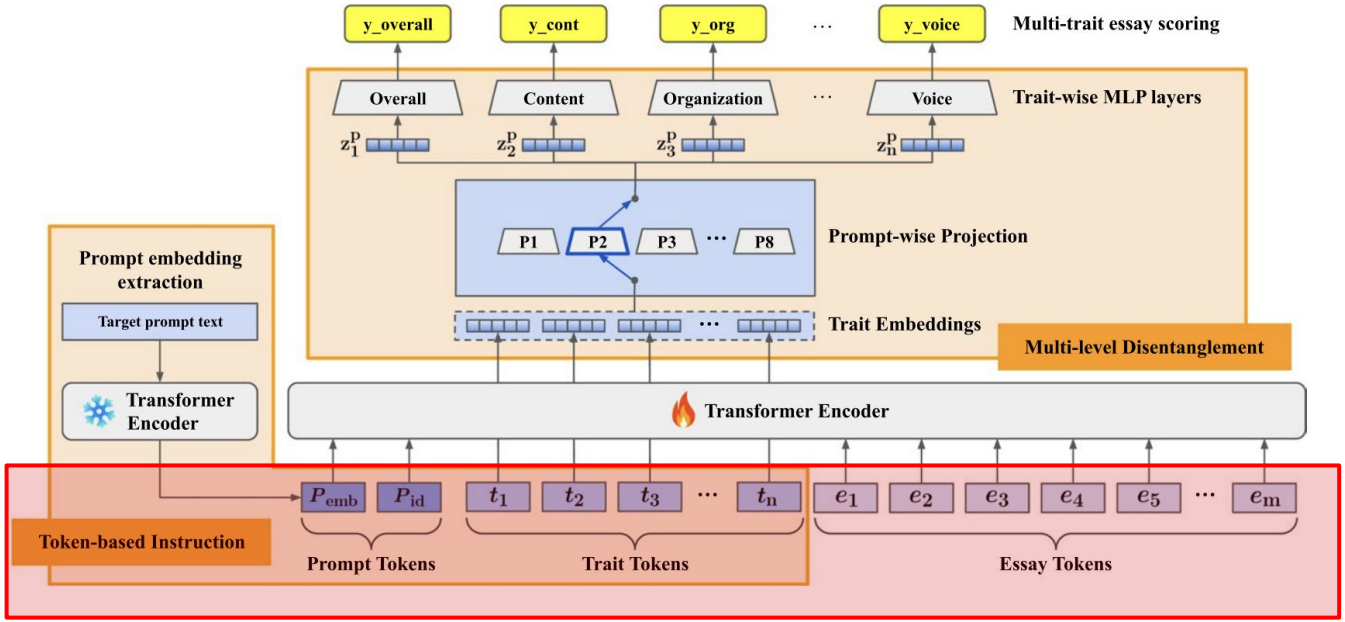


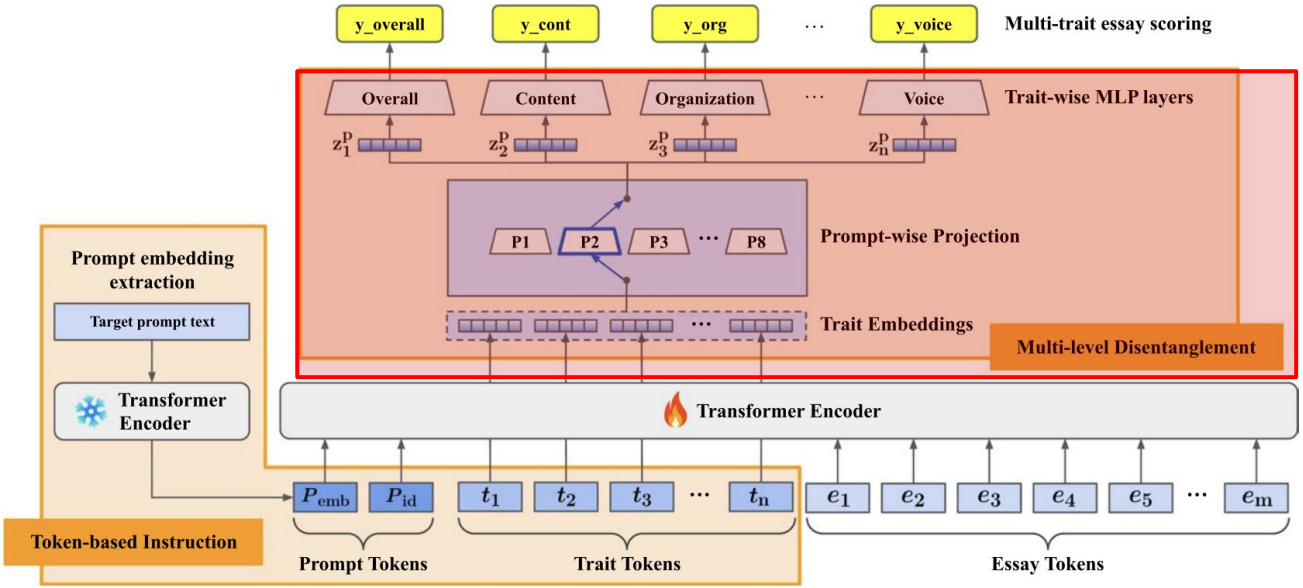
Fig. 3 Overview of MLPAS: Token-based instruction provides prompt-specific information about the target essay to guide the model in outputting the essay score for each trait token, while multi-level disentanglement dynamically adjusts the projection layers for fine-grained, multi-level disentanglement over prompts and traits

Token-based Instruction



$$S = [P^*|T|E] = [\underbrace{p_{\text{emb}}, p_{\text{id}}}_{\text{Prompt Tokens}}, \underbrace{t_1, t_2, \ldots, t_n}_{\text{Trait Tokens}}, \underbrace{e_1, e_2, \ldots, e_m}_{\text{Essay Tokens}}].$$

Multi-level Disentanglement



Experimental Setup

- Dataset
  - ASAP
  - ASAP++
  - Feedback Prize
- Evaluation Metric
  - Quadratic Weighted Kappa (QWK)
- Baseline
  - Traditional AES models
    - STL-LSTM, HISK, MTL-BiLSTM
  - Transformer-based model
    - ArTS (T5 encoder–decoder)
  - LLM-based methods
    - GPT-4, GPT-4o (zero-shot and few-shot)
  - Cross-prompt baselines
    - CTS, PMAES, PLAES, GAPS

Table 2 Overview of ASAP/ASAP++ dataset used in experiments

Prompt	Essay domain	# Essays	Avg. length	Available traits
1	Computer usage	1,785	350	Over, Cont, WC, Org, SF, Conv
2	Library censorship	1,800	350	Over, Cont, WC, Org, SF, Conv
3	Cyclist setting	1,726	150	Over, Cont, PA, Nar, Lang
4	Story analysis	1,772	150	Over, Cont, PA, Nar, Lang
5	Memoir mood	1,805	150	Over, Cont, PA, Nar, Lang
6	Empire state building	1,800	150	Over, Cont, PA, Nar, Lang
7	Patience story	1,569	150	Over, Cont, Org, Conv, Style
8	Laughter importance	723	150	Over, Cont, WC, Org, SF, Conv, Voice

Over: Overall, Cont: Content, WC: Word Choice, Org: Organization, SF: Sentence Fluency, Conv: Conventions, PA: Prompt Adherence, Nar: Narrativity, Lang: Language

Table 3 Overview of feedback prize dataset used in experiments

Prompt	Essay domain	# Essays	Avg. length	Available traits
1	Education and school life	1,095	431.4	Coh, Syn, Voc, Phr, Gram, Conv
2	Personal growth and attitude	1,865	435.9	Coh, Syn, Voc, Phr, Gram, Conv
3	Social experiences and future	945	418.8	Coh, Syn, Voc, Phr, Gram, Conv

Conv: Conventions, Gram: Grammar, Phr: Phraseology, Voc: Vocabulary, Syn: Syntax, Coh: Cohesion

Main Results – Overall Performance

Table 5 Trait-wise agreement with human ratings on the ASAP/ASAP++ dataset.

Model type	Model	Overall	Content	PA	Lang	Nar	Org	Conv	WC	SF	Style	Voice	AVG↑(SD↓)
	HISK	0.718	0.679	0.697	0.605	0.659	0.610	0.527	0.579	0.553	0.609	0.489	0.611 (-)
Non-transformer	STL-LSTM	0.750	0.707	0.731	0.640	0.699	0.649	0.605	0.621	0.612	0.659	0.544	0.656 (-)
	MTL-BiLSTM	0.764	0.685	0.701	0.604	0.668	0.615	0.560	0.615	0.598	0.632	0.582	0.638 (-)
Encoder-Decoder	ArTS	0.754	0.730	**0.751**	**0.698**	0.725	0.672	0.668	0.679	0.678	**0.721**	0.570	0.695 (±0.018)
Encoder-only	MLPAS **(ours)**	**0.771**	**0.746**	**0.751**	0.691	**0.732**	**0.710**	**0.701**	**0.701**	**0.699**	0.702	**0.650**	**0.714** (±0.007)

PA: Prompt Adherence, Lang: Language, Nar: Narrativity, Org: Organization, Conv: Conventions, WC: Word Choice, SF: Sentence Fluency

Table 6 Prompt-wise agreement with human ratings on the ASAP/ASAP++ dataset (P1-P8 denote the prompts)

Model type	Model	P1	P2	P3	P4	P5	P6	P7	P8	AVG↑(SD↓)
	HISK	0.674	0.586	0.651	0.681	0.693	0.709	0.641	0.516	0.644 (-)
Non-transformer	STL-LSTM	0.690	0.622	0.663	0.729	0.719	0.753	0.704	0.592	0.684 (-)
	MTL-BiLSTM	0.670	0.611	0.647	0.708	0.704	0.712	0.684	0.581	0.665 (-)
Encoder-Decoder	ArTS	0.708	0.706	**0.704**	0.767	0.723	**0.776**	0.749	0.603	0.717 (±0.025)
Encoder-only	MLPAS **(ours)**	**0.718**	**0.726**	0.703	**0.771**	**0.727**	0.765	**0.753**	**0.683**	**0.731** (±0.007)

Main Results – Overall Performance

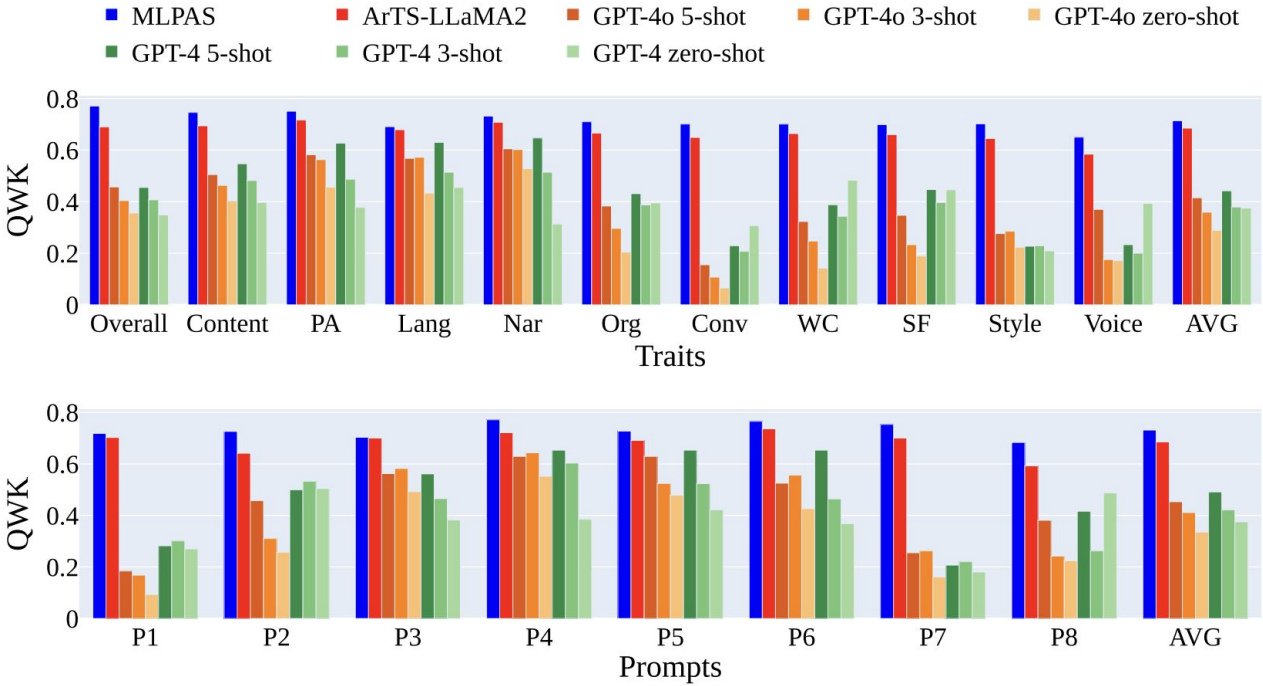


Fig. 4 Agreement with human ratings on the ASAP/ASAP++ dataset using LLMs as essay evaluators

Main Results – Generalization & Efficiency

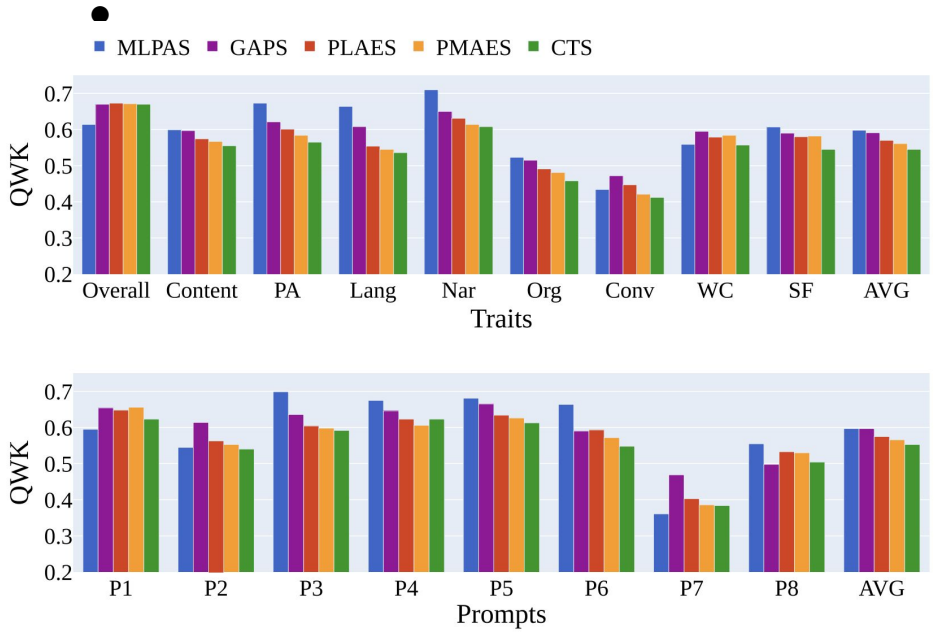
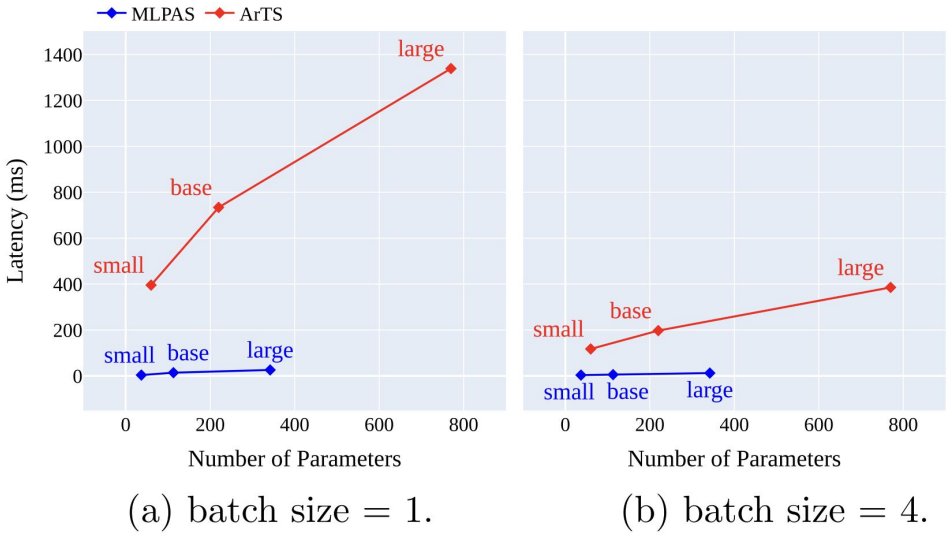


Fig. 5 Cross-prompt performance comparison on the ASAP/ASAP++ dataset



(a) batch size = 1. (b) batch size = 4.

Fig. 6 Trade-off between model size (# parameters) and latency (ms) when using two different batch sizes

In-depth Analysis

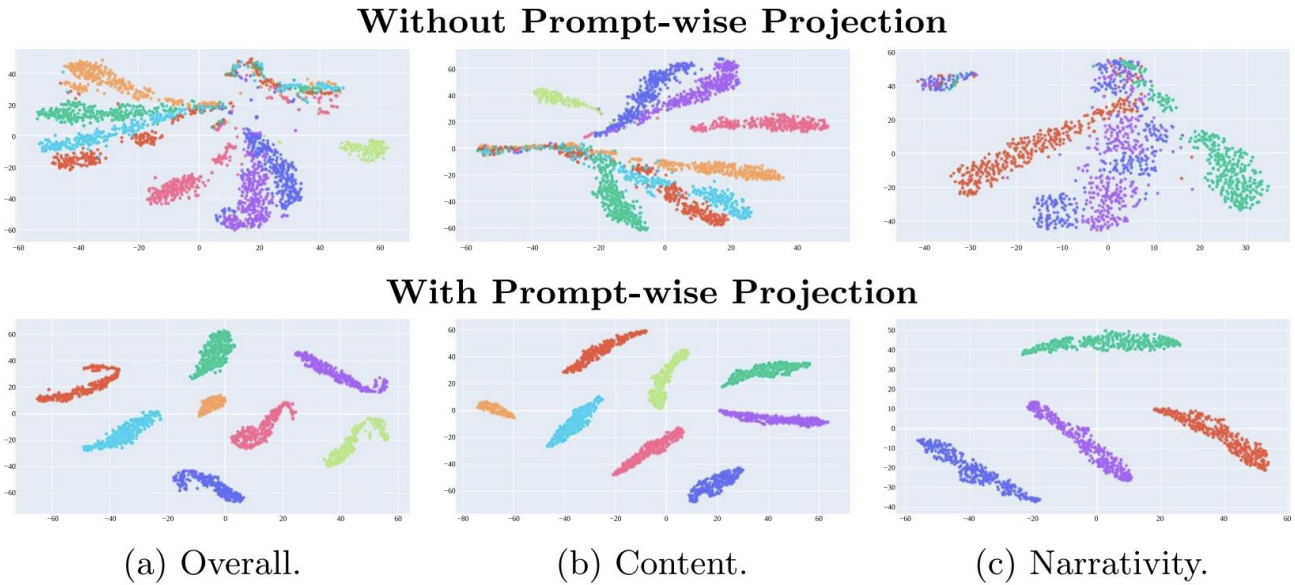


Fig. 7 T-SNE visualization of the traits with the same name across different prompts, with and without the prompt-wise projection of MLPAS, Trait embeddings just before regression are used, with each prompt color coded

In-depth Analysis

Table 10 Ablation study results based on traits

Model	Overall	Content	PA	Lang	Nar	Org	Conv	WC	SF	Style	Voice	AVG↑(SD↓)
MLPAS	**0.771**	**0.746**	0.751	0.691	**0.732**	**0.710**	**0.701**	**0.701**	**0.699**	**0.702**	**0.650**	**0.714** (±0.007)
(1) w/o essay prompt	0.759	0.735	**0.755**	0.690	0.728	0.690	0.698	0.695	0.689	0.693	0.622	0.705 (±0.004)
(2) w/o prompt ID token	0.768	0.740	**0.755**	0.692	0.727	0.700	0.694	0.698	0.692	0.698	0.607	0.706 (±0.009)
(3) w/o trait tokens	0.770	0.741	0.752	**0.695**	**0.732**	0.692	0.693	**0.701**	0.690	0.683	0.633	0.707 (±0.005)
(4) w/o prompt-wise projection	0.760	0.734	0.754	0.682	0.727	0.696	0.696	0.703	0.697	0.694	0.635	0.707 (±0.009)

Conclusion

- We proposed MLPAS, a multi-level disentanglement framework for multi-prompt essay scoring.
- MLPAS disentangles prompt- and trait-specific information using token-based instructions and prompt-wise projection.
- Our model achieves state-of-the-art performance, while being more efficient and generalizable than prior methods.
- This makes MLPAS a practical and powerful solution for real-world educational assessment.

Thank you

Q&A